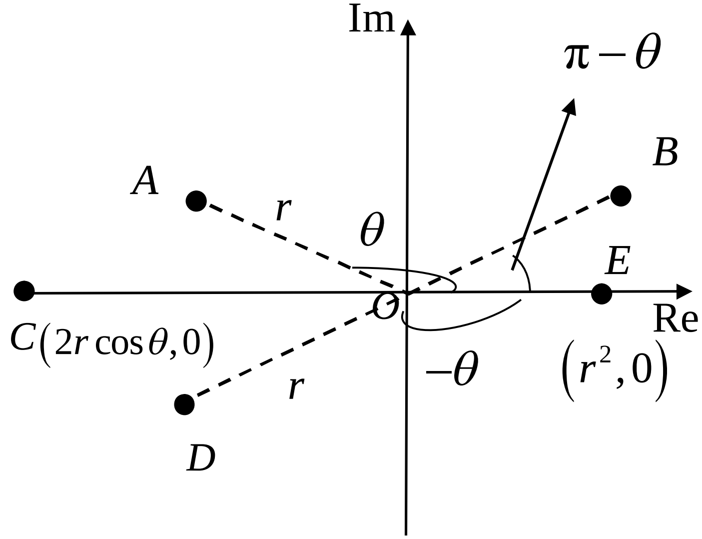
2019 NJC Preliminary Examination H2 Mathematics Paper 1 Marker's Comments

<table>
<tr><th>1</th><th>Suggested Solution</th><th>Comments</th></tr>
<tr><td>(i)</td><td>
</td><td>Most students are not able to figure out how to find $z-w$.

Method 1
Let $z=a+bi$
then $w=-a+bi$ (due to symmetry from diagram)
We know that the $z=re^{i\theta}=a+bi=r\cos\theta+i(r\sin\theta)$
So real component of z is $r\cos\theta$.
$z-w=a+bi-(-a+bi)=2a=2r\cos\theta$
Method 2
$z-w=re^{i\theta}-re^{i(\pi-\theta)}=r\left(e^{i\theta}-e^{i(\pi-\theta)}\right)$
$=r\left(e^{i\theta}-e^{i(-\theta)}e^{i\pi}\right)=r\left(e^{i\theta}-(-1)e^{i(-\theta)}\right)$
$=r\left(e^{i\theta}+e^{i(-\theta)}\right)=r(\cos\theta+i\sin\theta+\cos(-\theta)+i\sin(-\theta))$
$=r(\cos\theta+i\sin\theta+\cos\theta-i\sin\theta)$
$=2r\cos\theta$

Note that the lines joining O to the different points e.g. A, are dotted.

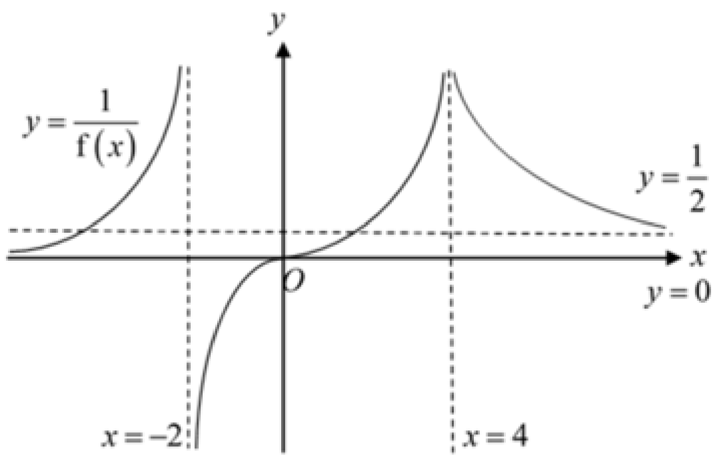
Points A and B need to be symmetrical about imaginary axis and point points A and D need to be symmetrical about the real axis and BOD should be collinear.</td></tr>
<tr><td>(ii)</td><td>ACDE is a kite.</td><td>Please note the geometrical properties of a kite e.g. two pairs of equal-length sides that are adjacent to each other.</td></tr>
</table>

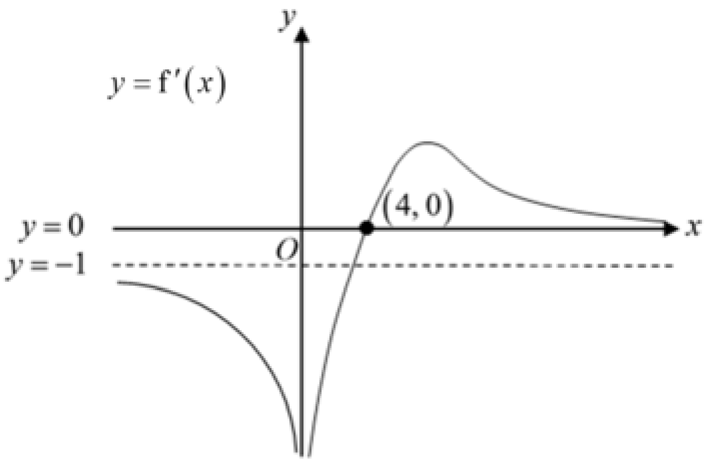
2	**Suggested Solution**	**Comments**
(i)	$\dfrac{2x^2+5x+9}{x-7} \le x-4$ $\dfrac{2x^2+5x+9}{x-7} - (x-4) \le 0$ $\dfrac{2x^2+5x+9-(x-4)(x-7)}{x-7} \le 0$ $\dfrac{2x^2+5x+9-x^2+11x-28}{x-7} \le 0$ $\dfrac{x^2+16x-19}{x-7} \le 0$ $\dfrac{(x+8)^2-83}{x-7} \le 0$ $\dfrac{(x+8+\sqrt{83})(x+8-\sqrt{83})}{x-7} \le 0$ $x \le -8-\sqrt{83}$ or $\sqrt{83}-8 \le x < 7$	Firstly, GC is no allowed for this question as per instruction. Do take note! Do not cross multiply, i.e. $2x^2+5x+9 \le (x-4)(x-7)$. One cannot be sure of the sign for $x-7$. $x \le -8-\sqrt{83}$ or $\sqrt{83}-8 \le x < 7$ cannot be the final answers. $x=7$ must be excluded because $\dfrac{2x^2+5x+9}{x-7}$ cannot admit 7!
(ii)	Replace x with e^{-x}, $\dfrac{2e^{-2x}+5e^{-x}+9}{e^{-x}-7} \le e^{-x}-4$ $\dfrac{2e^{-x}+5+9e^{x}}{1-7e^{x}} \le e^{-x}-4$ $\dfrac{2+5e^{x}+9e^{2x}}{7e^{x}-1} \ge 4e^{x}-1$ $e^{-x} \le -8-\sqrt{83}$ (reject) or $\sqrt{83}-8 \le e^{-x} < 7$ $\ln\left(\sqrt{83}-8\right) \le -x < \ln 7$ $-\ln 7 < x \le -\ln\left(\sqrt{83}-8\right)$	Let $x=e^{-x}$. This is very poor presentation. Use a different variable and go for $u=e^{-x}$ say. $e^{-x} \le -8-\sqrt{83}$ (reject) or $\sqrt{83}-8 \le e^{-x} < 7$ We are solving an equality so answer must be in terms of x.

3	Suggested Solution	Comments
(i)	By Ratio Theorem, $\overrightarrow{OS}=\dfrac{\lambda\mathbf{b}+(1-\lambda)\mathbf{a}}{\lambda+(1-\lambda)}=\mathbf{a}+\lambda(\mathbf{b}-\mathbf{a})$	Most students were able to apply the Ratio Theorem here.
(ii)	If the cargo ship is closest, $\overrightarrow{OS}\perp\overrightarrow{AB}$ and thus, $\overrightarrow{OS}\cdot\overrightarrow{AB}=0$. Consequently, $\overrightarrow{OS}\cdot\overrightarrow{AB}=(\mathbf{a}+\lambda(\mathbf{b}-\mathbf{a}))\cdot(\mathbf{b}-\mathbf{a})$ $0=\mathbf{a}\cdot(\mathbf{b}-\mathbf{a})+\lambda(\mathbf{b}-\mathbf{a})\cdot(\mathbf{b}-\mathbf{a})$ $\lambda=\dfrac{\mathbf{a}\cdot(\mathbf{a}-\mathbf{b})}{(\mathbf{b}-\mathbf{a})\cdot(\mathbf{b}-\mathbf{a})}$	Although students are generally able to identify that the vectors must be perpendicular and thus, $\overrightarrow{OS}\cdot\overrightarrow{AB}=0$. Manipulation of the vector algebra is generally convoluted as students expands the terms unnecessarily. Some students used incorrect vector notation for $\mathbf{b}\cdot\mathbf{b}$ and write $\mathbf{b}^2$, which is meaningless.
(iii)	$\overrightarrow{OS}=\mathbf{a}+\lambda(\mathbf{b}-\mathbf{a})$ $=\mathbf{a}+\dfrac{\mathbf{a}\cdot(\mathbf{a}-\mathbf{b})}{(\mathbf{b}-\mathbf{a})\cdot(\mathbf{b}-\mathbf{a})}(\mathbf{b}-\mathbf{a})$ $=\mathbf{a}+\dfrac{\mathbf{a}\cdot\mathbf{a}-\mathbf{a}\cdot\mathbf{b}}{\mathbf{b}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{a}-2\mathbf{a}\cdot\mathbf{b}}(\mathbf{b}-\mathbf{a})$ $=\mathbf{a}+\dfrac{\lvert\mathbf{a}\rvert^2-\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos(60^\circ)}{\lvert\mathbf{b}\rvert^2+\lvert\mathbf{a}\rvert^2-2\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos(60^\circ)}(\mathbf{b}-\mathbf{a})$ $=\mathbf{a}+\dfrac{9-(3)(2)\left(\frac{1}{2}\right)}{9+4-2(3)(2)\left(\frac{1}{2}\right)}(\mathbf{b}-\mathbf{a})$ $=\mathbf{a}+\dfrac{6}{7}(\mathbf{b}-\mathbf{a})$ $=\dfrac{1}{7}\mathbf{a}+\dfrac{6}{7}\mathbf{b}$	

4	Suggested Solution	Comments
(i)	$\frac{p}{1-x}+\frac{q}{(2+x)^2}=\frac{p(2+x)^2+q(1-x)}{(1-x)(2+x)^2}$ $=\frac{4p+4px+px^2+q-qx}{(1-x)(2+x)^2}$ $=\frac{px^2+(4p-q)x+(4p+q)}{(1-x)(2+x)^2}$	Many students tried to do partial fraction method which is a waste of time. Note that you can show from left to right or vice versa.
(ii)	$\frac{px^2+(4p-q)x+(4p+q)}{(1-x)(2+x)^2}+r\ln(1-3x)$ $=\frac{p}{1-x}+\frac{q}{(2+x)^2}+r\ln(1-3x)$ $=p(1-x)^{-1}+q\left[2^{-2}\left(1+\frac{x}{2}\right)^{-2}\right]+r\ln(1-3x)$ $=p(1+x+x^2+\ldots)$ $+\frac{1}{4}q\left[1+(-2)\left(\frac{x}{2}\right)+\frac{(-2)(-3)}{1\cdot 2}\left(\frac{x}{2}\right)^2+\ldots\right]$ $+r\left[-3x-\frac{(-3x)^2}{2}+\ldots\right]$ $\approx p(1+x+x^2)+\frac{1}{4}q\left(1-x+\frac{3}{4}x^2\right)+r\left(-3x-\frac{9}{2}x^2\right)$ $=\left(p+\frac{1}{4}q\right)+\left(p-\frac{1}{4}q-3r\right)x+\left(p+\frac{3}{16}q-\frac{9}{2}r\right)x^2$ $\equiv 4-4x+x^2$ Comparing coefficients,	Note the factoring of 2 out gives 2^{-2} and there is no need for maclaurin's series expansion – juse use the Binomial series expansion. Note the $\frac{x}{2}$ in $\left(1+\frac{x}{2}\right)^{-2}$.

	$\begin{cases} p+\frac{1}{4}q+0r=11 \\ p-\frac{1}{4}q-3r=-3 \\ p+\frac{3}{16}q-\frac{9}{2}c=1 \end{cases}$ By GC, $p=7, q=16, r=2$	
(iii)	$\lvert -x\rvert<1$ and $\left\lvert\frac{x}{2}\right\rvert<1$ and $-1<-3x\le 1$ Thus $-\frac{1}{3}\le x<\frac{1}{3}$	Please present in such a way to show the marker that you are considering the intersection of values in three inequalities as well as the use of 'and'.

5	**Suggested Solution**	**Comments**
(i)		Generally well done. **Common mistakes:** 1. Graph did not approach the x-axis as $x\to-\infty$. 2. Did not label $y=0$. 3. Did not label the origin / x/y intercept.

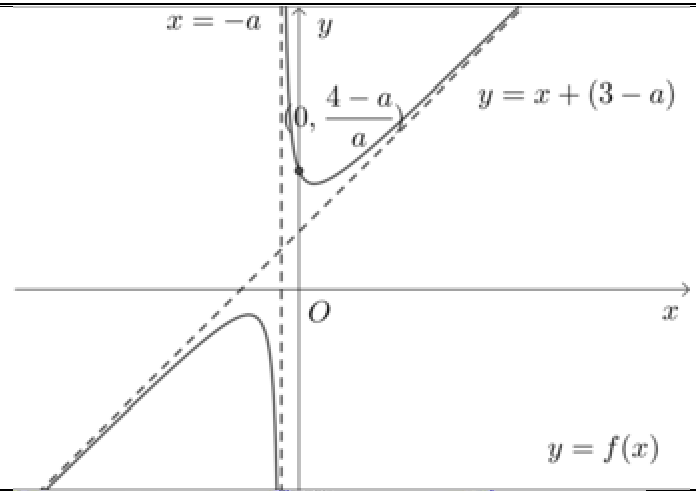
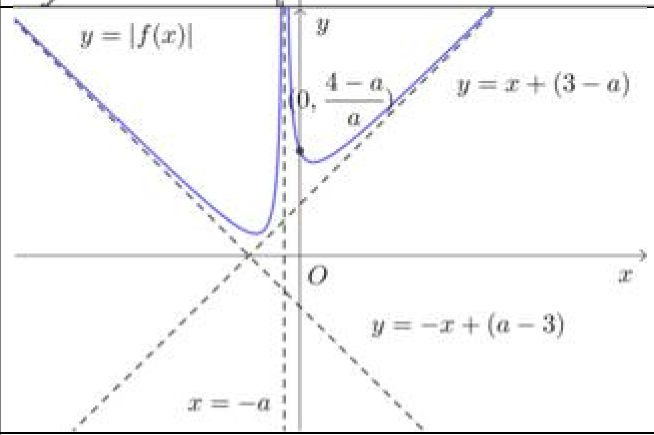
(ii)		Generally well done. **Common mistakes:** 1. Graph did not approach $y = -1$ as $x \to -\infty$. 2. Did not label $x = 0$ and $y = 0$. 3. Curve not smooth (weird curvature) at (4, 0).
	$f(x) = 2 - x$ Replace x by $x - 2$ $f(x-2) = 4 - x$ Replace x by $2x$ $f(2x-2) = 4 - 2x$ Hence $a = 2,\ b = -2$.	Badly done. Most students are unable to identify the correct replacement technique to transform the equation of the asymptote.

6	Suggested Solution	Comments
(a)	$\left\lvert\overrightarrow{OA}\right\rvert^2 = 6^2 + t^2 + 10^2 = 136 + t^2$ $\cos\theta = \frac{6}{\sqrt{136+t^2}} = \frac{3}{\sqrt{35}}$ $\frac{36}{136+t^2} = \frac{9}{35}$ $1260 = 1224 + 9t^2$ $9t^2 = 36$ $t = \pm 2$ $t = -2 \quad (\because t < 0)$	You must be familiar with the definition and formula of Direction Cosine. If point A has a position vector $\overrightarrow{OA}$, then the direction cosine along the x, y and z-axis is given by $\frac{\overrightarrow{OA}}{\left\lvert\overrightarrow{OA}\right\rvert}\cdot\mathbf{i}$, $\frac{\overrightarrow{OA}}{\left\lvert\overrightarrow{OA}\right\rvert}\cdot\mathbf{j}$, $\frac{\overrightarrow{OA}}{\left\lvert\overrightarrow{OA}\right\rvert}\cdot\mathbf{k}$. $\cos\theta$ is positive so θ acute with θ being the angle between the vector and the x-axis.

	Angle with the *yz*-plane is $90° - \cos^{-1}\frac{3}{\sqrt{35}} \approx 30.5°$	With some visualisation, one can see that the angle of the *yz* plane is $90° - \cos^{-1}\theta$. Although the question says deduce, some of you computed the angle of a line and a plane using existing methods.
(b) (i)	$(4, 0, 0)$ is a point on p_1, $(0, 0, -k)$ is a point on p_2. The midpoint $(2, 0, -0.5k)$ is on *p*. A normal vector of *p* is $\begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$, so a vector equation of *p* is $\mathbf{r} \cdot \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ -0.5k \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = 4 - 0.5k$ i.e. $\mathbf{r} \cdot \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = 4 - 0.5k$	There are many approaches to this • Using ratio theorem • Using the distance between plane and origin • Using the distance between 2 planes
(b) (ii)	Let *N* be the point of intersection of *l* and *p*. $\frac{x-1}{3} = \frac{2y-1}{18} = \frac{3-z}{1} \Rightarrow \frac{x-1}{3} = \frac{y-0.5}{9} = \frac{z-3}{-1}$ Vector equation of *l*: $\mathbf{r} = \begin{pmatrix} 1 \\ 0.5 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 9 \\ -1 \end{pmatrix}$, $\lambda \in \mathbb{R}$. Vector equation of *p*: $\mathbf{r} \cdot \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = 4 - 0.5(13) = -2.5$	Part (ii) is dependent on (i). If the equation of *p* is found in (ii), a point that is common on *l* and *p* will be equidistant between p_1 and p_2. However, the question can still be solved if the equation of *p* is not found i.e. by considering the distance between *l* with p_1 and *l* with p_2. Since *l* is not parallel to either p_1 or p_2, we can find a point on *l* such that it is equidistance to p_1 and p_2.

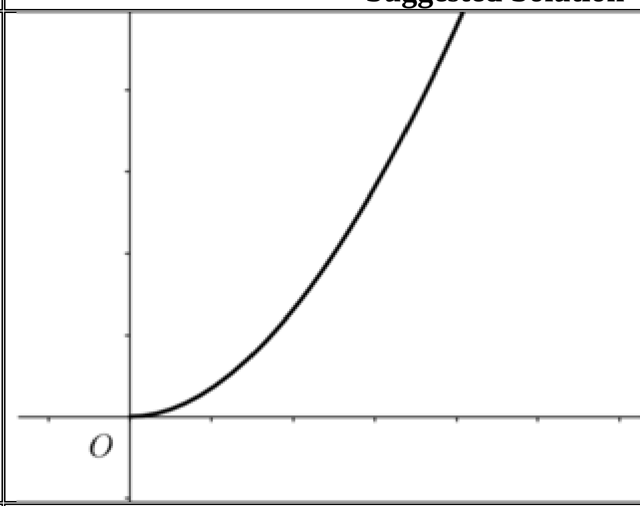
	Let $\overrightarrow{ON}=\begin{pmatrix}1+3\lambda\\0.5+9\lambda\\3-\lambda\end{pmatrix}$ for some $\lambda\in\mathbb{R}$. $\begin{pmatrix}1+3\lambda\\0.5+9\lambda\\3-\lambda\end{pmatrix}\cdot\begin{pmatrix}2\\-2\\1\end{pmatrix}=-2.5$ $2+6\lambda-1-18\lambda+3-\lambda=-2.5$ $-13\lambda=-6.5$ $\lambda=0.5$ $\overrightarrow{ON}=\begin{pmatrix}1+3(0.5)\\0.5+9(0.5)\\3-0.5\end{pmatrix}$ The coordinates are $(2.5, 5, 2.5)$	

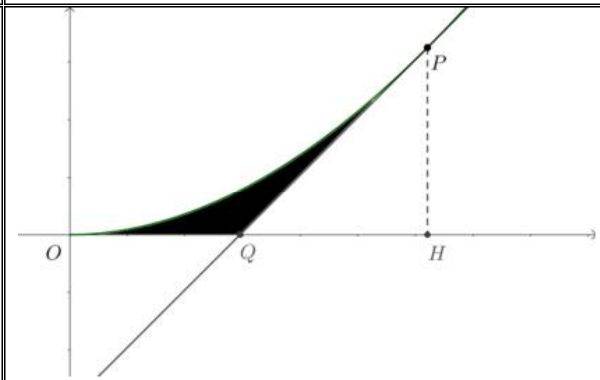
7	**Suggested Solution**	**Comments**
(i)	$\mathrm{f}(x)=\dfrac{x^2+3x+4-a}{x+a}=x+(3-a)+\dfrac{a^2-4a+4}{x+a}$ The asymptotes are $x=-a$ and $y=x+(3-a)$ For *y*-intercept, let $x=0$. Then $y=\dfrac{4-a}{a}$. For x-intercept, let $y=0$. Then $x^2+3x+4-a=0$. But $3^2-4(4-a)=9-16+4a=4a-7<0$ as $0<a<1.5$, there is no solution to the equation thus the graph has no *x*-intercept.	Students need to master long division with unknowns at the denominator. Students are to recognise that f is a rational function consisting of a vertical and an oblique asymptote. **Common mistake:** 1. Coordinates of intercept not labelled.

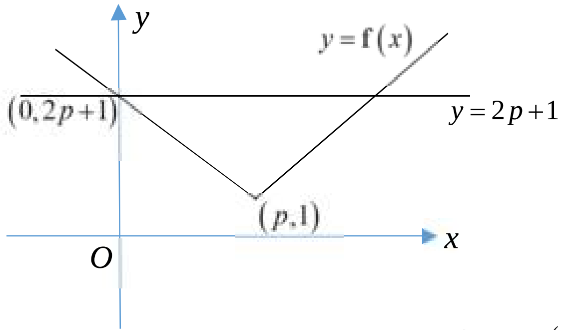
		
		Almost all students sketched the graph for $y=\lvert \mathrm{f}(x)\rvert$ correctly. **Common mistakes**: 1. Equation of the reflected asymptote found wrongly. 2. The reflected asymptote should be a reflection of the original oblique asymptote about the *x*-axis.
(iii)	<u>Method 1</u> $\mathrm{f}(x)=x+(3-a)+\dfrac{a^2-4a+4}{x+a}$ $\qquad=x+(3-a)+\dfrac{(a-2)^2}{x+a}$ At stationary points, $\mathrm{f}'(x)=1-\dfrac{(a-2)^2}{(x+a)^2}=0$	From the graph in **(ii)**, students should differentiate and find the stationary points to determine the range of values of *m*. However, many students differentiated wrongly, likely due to error in long division in the first part of the question.

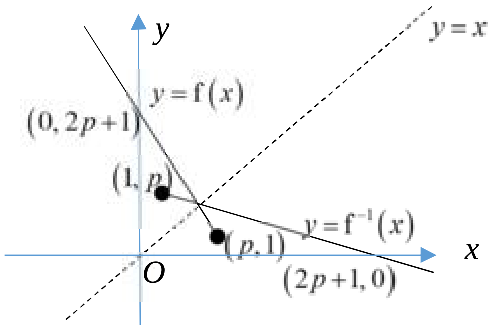
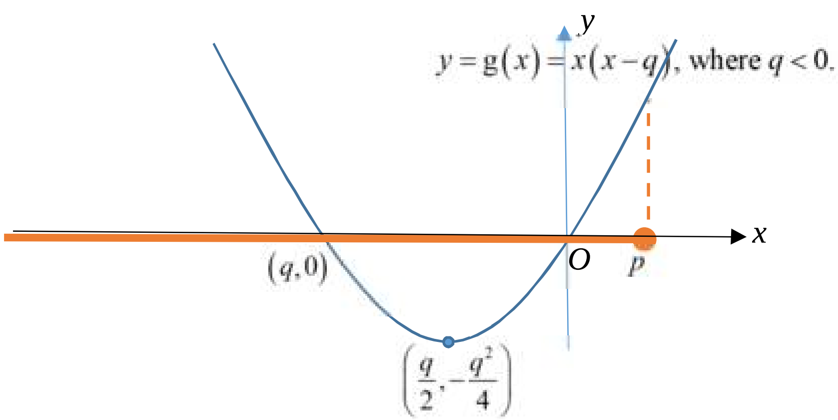
	$(x+a)^2=(a-2)^2$ $x+a=a-2$ or $x+a=-(a-2)$ $x=-2$ or $x=2-2a$ Correspondingly, the y-coordinates of the points: $\mathrm{f}(-2)=-2+(3-a)+\dfrac{(a-2)^2}{a-2}=1-a+a-2=-1$ $\mathrm{f}(2-2a)=(2-2a)+(3-a)+\dfrac{(a-2)^2}{(2-2a)+a}$ $=5-3a+\dfrac{(a-2)^2}{2-a}$ $=5-3a+2-a$ $=7-4a$ On $y=\lvert\mathrm{f}(x)\rvert$, the stationary points are $(-2,1)$ and $(2-2a,7-4a)$. Since $0<a<\dfrac{3}{2}$ $\therefore 7-4a>1$. Hence the line $y=m$ cuts the graph of $y=\lvert\mathrm{f}(x)\rvert$ at two distinct points when $1<m<7-4a$. <u>Method 2</u> Consider the equation $\mathrm{f}(x)=m$ to have no real root $\dfrac{x^2+3x+4-a}{x+a}=m$ $x^2+3x+4-a=mx+am$ $x^2+(3-m)x+(4-a-am)=0$	Most students used Method 2. However, students are not sure what to do with the modulus sign. They should realise that the modulus can be removed to simplify their calculation.

	$(3-m)^2-4(4-a-am)$ $=m^2-6m+9-16+4a+4am$ $=m^2+(4a-6)m+(4a-7)$ $=(m-(-1))[m-(7-4a)]<0$ $-1<m<7-4a$ Referring to the graph of $y=\mathrm{f}(x)$ and $y=\lvert\mathrm{f}(x)\rvert$, since $0<a<\dfrac{3}{2}$ $\therefore 7-4a>1$. We can deduce that $1<m<7-4a$ for $\lvert\mathrm{f}(x)\rvert=m$ to have exactly two distinct roots.	

8	**Suggested Solution**	**Comments**
(i)	 	This part is not done well by some of the students. **Common mistakes:** 1. Forget to set the values of t in GC. 2. Fail to show the feature that the only x-intercept of the curve is at the origin by making the curve lie on the x-axis for a segment. Students need to remember to adjust the window setting in GC to 1. sketch any parametric curve, 2. display key features clearly on the screen.
(ii)	$x=t^2-4t,\quad y=3t^2-t^3,\quad t\le 0$ $\dfrac{\mathrm{d}x}{\mathrm{d}t}=2t-4, \dfrac{\mathrm{d}y}{\mathrm{d}t}=6t-3t^2, \dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{6t-3t^2}{2t-4}=\dfrac{-3t(t-2)}{2(t-2)}=-\dfrac{3t}{2}$	This part is well done by most of the students.

	At P, the equation of tangent is $y-\left(3p^2-p^3\right)=-\frac{3}{2}p\left[x-\left(p^2-4p\right)\right]$ $y=-\frac{3}{2}px+\frac{3}{2}p^3-6p^2+3p^2-p^3$ $y=-\frac{3}{2}px+\frac{1}{2}p^3-3p^2$	
(iii)	From the sketch of C, the gradient of the tangent must be 1. $-\frac{3}{2}p=1 \Rightarrow p=-\frac{2}{3}$. The coordinates of P are $\left(\frac{28}{9},\frac{44}{27}\right)$	This part is not done well by many students. **Common mistakes:** 1. Wrongly assume the ratio of y to x is 1 for the gradient of the tangent. 2. Fail to make use the result of part **(ii)**. 3. Forget to find the coordinates after finding the parameter value.
	 Area $=\int_0^{\frac{28}{9}} y\,\mathrm{d}x-$ area of triangle PQH $=\int_0^{-\frac{2}{3}}\left(3t^2-t^3\right)\left(2t-4\right)\mathrm{d}t-\frac{1}{2}\left(\frac{44}{27}\right)\left(\frac{44}{27}\right)$ $\approx 1.731687-1.327846$ $=0.403841$ ≈ 0.404	This part is poorly done. **Common mistakes:** 1. See the region as simply an area under the parametric curve. 2. See the region as an area between the curve and the line over the same interval 3. Cannot recall the formula to find the area under a parametric curve. 4. Make mistakes in calculating the height or base of the triangle. In addition, whenever a question does not specify 'exact' or 'without using a calculator', it is encouraged to use a GC to evaluate the definite integral numerically, to save time.

9	**Suggested Solution**	**Comments**
(i)	 The line $y=2p+1$ cuts the graph of $y=\mathrm{f}(x)$ twice. Hence f is not one-one function and its inverse does not exist. OR The line $y=l$, where $l>1$, cuts the graph of $y=\mathrm{f}(x)$ twice. Hence f is not one-one function and its inverse does not exist.	Students need to be mindful that when using graphical method to show whether f is one-one function, they need to sketch the graph of f and the horizontal line on the same diagram. **Common Mistakes** 1. Did not label the important feature of the graph of f. In this case, the vertex and the axial intercept of f. 2. Did not sketch the mentioned horizontal line on the same diagram as graph of f. 3. Did not state the range of l where $y=l$ cuts the graph of f at least once.
(ii)	Largest value of k is p. Based on restricted domain of f i.e. $x \le p$, such that inverse of f exists, $y=2\lvert x-p\rvert+1$ $\Rightarrow y=-2(x-p)+1$ $\Rightarrow y-1=-2(x-p)$ $\Rightarrow x-p=\dfrac{1-y}{2}$ $\therefore x=p+\dfrac{1-y}{2}$ $\therefore \mathrm{f}^{-1}(x)=p+\dfrac{1-x}{2}$ $\therefore \mathrm{D}_{\mathrm{f}^{-1}}=\mathrm{R}_{\mathrm{f}}=[1,\infty)$	Need to learn to choose the right expression for f based on the restricted domain of f. $\lvert x-p\rvert=\begin{cases}(x-p), \text{ where } (x-p)\ge 0 \Leftrightarrow x\ge p \\ -(x-p), \text{where } (x-p)<0 \Leftrightarrow x<p\end{cases}$

(iii)		Graph of f and inverse of f must be symmetrical about the line $y = x$. **Common Mistakes:** 1. Did not sketch the line of reflection $y = x$ 2. Not aware that image of of (x, y) reflected about the line $y = x$ gives (y, x).
	$\mathrm{f}(x)=\mathrm{f}^{-1}(x)$ $\Rightarrow \mathrm{f}^{-1}(x)=x$ $\Rightarrow p+\dfrac{1-x}{2}=x$ $\Rightarrow 2p+1-x=2x$ $\Rightarrow 3x=2p+1$ $\therefore x=\dfrac{2p+1}{3}$ OR $\mathrm{f}(x)=\mathrm{f}^{-1}(x)$ $\Rightarrow \mathrm{f}(x)=x$ $\Rightarrow -2(x-p)+1=x$ $\Rightarrow -2x+2p+1=x$ $\Rightarrow 3x=2p+1$ $\therefore x=\dfrac{2p+1}{3}$	From the diagram in (iii), students are to observe that the point(s) of intersection of $\mathrm{f}(x)=\mathrm{f}^{-1}(x)$ lie on the line $y = x$. **Common Mistakes**: 1. Students did not choose the correct expression for f when solving the equation $\mathrm{f}(x)=x$.
(iv)	 $\mathrm{D}_{\mathrm{gf}^{-1}}=\mathrm{D}_{\mathrm{f}^{-1}}=[1,\infty)\xrightarrow[\mathrm{f}^{-1}]{}(-\infty,p]\xrightarrow[\text{into graph of g}]{}\left[-\dfrac{q^2}{4},\infty\right)=\mathrm{R}_{\mathrm{gf}^{-1}}$	Students are encouraged to use the mapping method to find the range of the composite function gf^{-1} and thus it is important to obtain the correct graph of g. **Common Mistakes**: 1. Students are not able to determine the coordinates of turning point of quadratic curve correctly. 2. Many assume that $\mathrm{R}_{\mathrm{gf}^{-1}}=\mathrm{R}_{\mathrm{g}}$ without verifying whether $\mathrm{R}_{\mathrm{f}^{-1}}=\mathrm{D}_{\mathrm{g}}$. That is to say students must verify that $\mathrm{R}_{\mathrm{f}^{-1}}=\mathrm{D}_{\mathrm{g}}$ is indeed true before applying $\mathrm{R}_{\mathrm{gf}^{-1}}=\mathrm{R}_{\mathrm{g}}$.

	Thus $R_{gf^{-1}} = \left[-\frac{q^2}{4}, \infty\right)$. Note: $D_{f^{-1}} = R_f = [1, \infty)$ and $R_{f^{-1}} = D_f = (-\infty, p]$	

10	Suggested Solution	Comments
(i)	Total depth $= 4.5 + 4.5\left(1-\frac{r}{100}\right) + 4.5\left(1-\frac{r}{100}\right)^2 + \ldots + 4.5\left(1-\frac{r}{100}\right)^9$ $= \dfrac{4.5\left[1-\left(1-\frac{r}{100}\right)^{10}\right]}{1-\left(1-\frac{r}{100}\right)}$ $= \frac{450}{r}\left[1-\left(1-\frac{r}{100}\right)^{10}\right]$	A lot of students did not understand the phrase "reduced by *r* %" and used $\frac{r}{100}$ as the common ratio.
(ii)	$\frac{450}{r}\left[1-\left(1-\frac{r}{100}\right)^{10}\right] = \frac{10}{2}\left[2(4.7)+9(-0.1)\right] = 42.5$ By GC, $r = 1.28$.	Some students failed to recognise that the value of the common difference is –0.1, not 0.1 when computing the total depth that the STK machine can drill. Sad to say, some students tried to solve the equation manually or manipulated the equation wrongly and then used GC to solve. This shows a lack of awareness of how to use the GC.
(iii)	Let the number of hours the machine has drilled be *n*. To satisfy requirement, $4.5(0.99)^{n-1} \geq 4.0$ From GC, $n \leq 12$ (or $n \leq 12.7$ algebraically) It should operate 12 complete hours.	Many students tried to solve $4.5(0.99)^n \geq 4.0$ while thinking that $4.5(0.99)^n$ gives the depth of the hole drilled in the *n*-th hour, not realising that this contradicts what they have written in **(i)**. Some students used a complicated method, which is to use the formula in **(i)**.
(iv)	For the alternative manufacturer's claim to be false, $\dfrac{4.5}{1-\left(1-\frac{r}{100}\right)} < 360 \quad r > 1.25$	While most students were able to recognise that sum of infinity must be used, they gave the common ratio as *r* or $1-r$, instead of $1-\frac{r}{100}$. Some of them used the correct common ratio in **(i)** correct but used the wrong one here.

(v)	<u>Method 1</u> For the STK machine, it can only drill at most 47 hours, as it will drill $4.7+(48-1)(-0.1)=0$ m in the 48th hour. Total depth drilled is $\frac{47}{2}\left[2\times 4.7+(47-1)(-0.1)\right]=112.8$. Therefore, it is not possible to drill a hole to 150 m deep. <u>Method 2</u> The total distance drilled by the STK machine in *m* hours is $\frac{m}{2}\left[2\times 4.7+(m-1)(-0.1)\right]=4.75m-0.05m^2$. Let $4.75m-0.05m^2=150$ $4.75m-0.05m^2=150$ $0.05m^2-4.75m+150=0$ $\text{discriminant}=4.75^2-4\times 150\times 0.05=-7.4375<0$ There is no real solution for *m*, so it is not possible to drill a hole to 150 m deep.	Equal number of students attempted both methods. The errors usually occur when computing the maximum total depth or manipulating the equation. Some students tried to use "sum to infinity", not realising that this does not make sense because the sum of infinity does not exist for an AP. Moreover the common difference is negative, so in this context it means that the machine can only drill up to a certain extent, so it is pointless to talk about drilling "forever".

11	**Suggested Solution**	**Comments**
(i)	$V=\frac{1}{3}\pi r^2 h \Rightarrow h=\frac{3V}{\pi r^2}$ $S=\pi r\sqrt{h^2+r^2}$ $=\pi r\sqrt{\left(\frac{3V}{\pi r^2}\right)^2+r^2}$ $=\pi r\sqrt{\frac{9V^2+\pi^2 r^6}{\pi^2 r^4}}$ $=\frac{\pi r}{\pi r^2}\sqrt{9V^2+\pi^2 r^6}$ $S^2=\frac{9V^2+\pi^2 r^6}{r^2}$ (Shown)	Well-done except for a handful who mistook *V* and *S* to include the cylindrical body.

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<td>(ii)</td>
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$S^2 = \dfrac{9V^2 + \pi^2 r^6}{r^2} = 9V^2 r^{-2} + \pi^2 r^4$
or let $y = 9V^2 r^{-2} + \pi^2 r^4$
Differentiating w.r.t. r,
$2S\dfrac{\mathrm{d}S}{\mathrm{d}r} = 9V^2\left(-2r^{-3}\right) + 4\pi^2 r^3$ ---------- (1)
$\dfrac{\mathrm{d}y}{\mathrm{d}r} = -18V^2 r^{-3} + 4\pi^2 r^3$
For $\dfrac{\mathrm{d}S}{\mathrm{d}r} = 0$,
$9V^2\left(-2r^{-3}\right) + 4\pi^2 r^3 = 0$
$\Rightarrow -18V^2 + 4\pi^2 r^6 = 0$
$\Rightarrow r = \sqrt[6]{\dfrac{9V^2}{2\pi^2}}$
Differentiating (1) w.r.t. r,
$2S\dfrac{\mathrm{d}^2S}{\mathrm{d}r^2} + 2\left(\dfrac{\mathrm{d}S}{\mathrm{d}r}\right)^2 = 9V^2\left(6r^{-4}\right) + 12\pi^2 r^2$
Or
$\dfrac{\mathrm{d}^2y}{\mathrm{d}r^2} = 54V^2 r^{-4} + 12\pi^2 r^2 > 0$
When $r = \sqrt[6]{\dfrac{9V^2}{2\pi^2}}$ and $\dfrac{\mathrm{d}S}{\mathrm{d}r} = 0$,
$\dfrac{\mathrm{d}^2S}{\mathrm{d}r^2} = \dfrac{1}{2S}\left(\dfrac{3V^2}{2r^4} + 12\pi^2 r^2\right) > 0.$
Thus S is minimum when $r = \sqrt[6]{\dfrac{9V^2}{2\pi^2}}$.
$\dfrac{h}{r} = \left(\dfrac{3V}{\pi r^2}\right)\left(\dfrac{1}{r}\right) = \dfrac{3V}{\pi r^3}$
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Many students were still not proficient in doing implicit differentiation when they tried to make S the subject by taking square root of the RHS, before finding $\dfrac{\mathrm{d}S}{\mathrm{d}r}$.

Many students were also unaware that V is a constant and hence $\dfrac{\mathrm{d}}{\mathrm{d}r}(V) = 0$ which simplifies subsequent workings.

Most students attempted to check for minimum by using the first derivative test but failed to realise that the test failed in this context because of the unknown numerical
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	When $r=\sqrt[6]{\frac{9V^2}{2\pi^2}}$, $\frac{h}{r}=\frac{3V}{\pi}\left(\sqrt[6]{\frac{2\pi^2}{9V^2}}\right)^3$ $=\frac{3V}{\pi}\left(\sqrt{\frac{2\pi^2}{9V^2}}\right)$ $=\frac{3V}{\pi}\left(\frac{\sqrt{2}\pi}{3V}\right)$ $=\sqrt{2}$ $\therefore h:r=\sqrt{2}:1$	value of r and hence we were not able to find the numerical values of the first derivative around the neighbourhood of the minimum point that explained the gradient of the tangents. Many students did not leave their answer in ratio, as required by the question.
(iii)	$(AB)\cos\frac{\pi}{3}=3\Rightarrow AB=3(2)=6$ $\angle ACB+\theta=\frac{\pi}{6}\Rightarrow\angle ACB=\frac{\pi}{6}-\theta$ and $\angle ABC=\pi-\frac{\pi}{6}=\frac{5\pi}{6}$ By sine rule, $\frac{AC}{\sin\frac{5\pi}{6}}=\frac{AB}{\sin\left(\frac{\pi}{6}-\theta\right)}$ $\Rightarrow AC=\frac{6\times\frac{1}{2}}{\sin\left(\frac{\pi}{6}-\theta\right)}=\frac{3}{\sin\left(\frac{\pi}{6}-\theta\right)}$ (Shown)	The method to answer this question is not unique and the provided solution uses the sine rule. Some students easily used the sine ratio on the right angle triangle *ANC*.

(iv)	$AC = \dfrac{3}{\sin\left(\frac{\pi}{6}-\theta\right)}$ $= \dfrac{3}{\sin\frac{\pi}{6}\cos\theta - \cos\frac{\pi}{6}\sin\theta}$ $\approx \dfrac{3}{\frac{1}{2}\left(1-\frac{1}{2}\theta^2\right)-\frac{\sqrt{3}}{2}\theta}$ $= 6\left[1-\left(\sqrt{3}\theta+\frac{1}{2}\theta^2\right)\right]^{-1}$ $\approx 6\left[1+\left(\sqrt{3}\theta+\frac{1}{2}\theta^2\right)+\left(\sqrt{3}\theta+\frac{1}{2}\theta^2\right)^2\right]$ $\approx 6\left[1+\left(\sqrt{3}\theta+\frac{1}{2}\theta^2\right)+\left(\sqrt{3}\theta\right)^2\right]$ $= 6+6\sqrt{3}\theta+3\theta^2+18\theta^2$ $= 6+6\sqrt{3}\theta+21\theta^2$ The length (in metres) to be cut off is $AC - AB \approx 6+6\sqrt{3}\theta+21\theta^2-6 = 6\sqrt{3}\theta+21\theta^2$	Most students failed to go beyond the approximations using small angles on $\sin\theta$ and $\cos\theta$. Common mistake includes writing '$\sin\left(\frac{\pi}{6}-\theta\right) \approx \frac{\pi}{6}-\theta$'. Those who attempted to do Maclaurin series increased the complexity of the solution that led to numerical mistakes and hence the final answer.